

HAT-P-18b

R-2151

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_250430 · created 2026-08-12T01:09:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460795.88553 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.036
Transit depth (Rp/Rs)^2	1.34 +/- 0.84 [%]
Orbital Inclination (inc)	89.98 +/- 1.65
Airmass coefficient 1 (a1)	0.964 +/- 0.014
Airmass coefficient 2 (a2)	0.034 +/- 0.051
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#3 - [342, 322]
Optimal Aperture	3.87
Optimal Annulus	4.59
Transit Duration (day)	0.105 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.116 ± 0.036 vs 0.1356 (deviation 0.52σ)
Duration fit vs published	0.105 d vs 0.1129 d (deviation 0.38σ)
Mid-time O–C	-2.4 min
Depth SNR	3.1
Residual scatter	1.44 %

Light curve

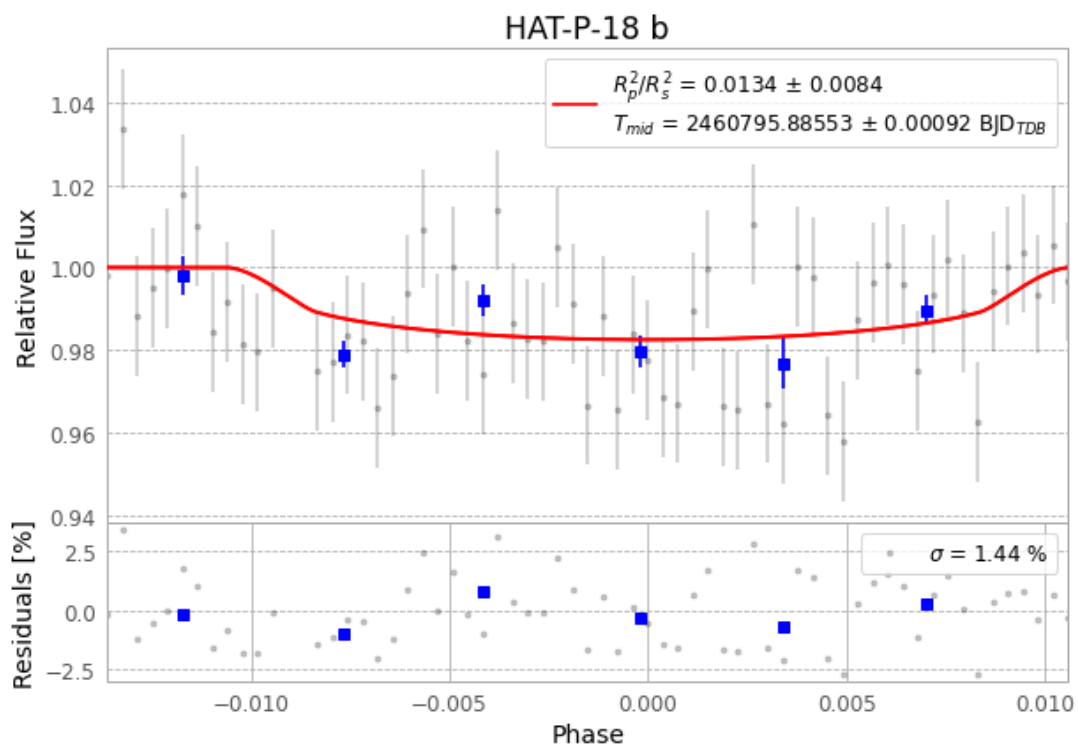


Figure 1 — FinalLightCurve_HAT-P-18 b_2025-04-30.png (SHA-256 24626331101a1f5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c721d4ad1d8104c2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

65 FITS frames (43 MB), HATP-18250430072116.FITS → HATP-18250430103317.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-250430.log (4bb2a11979bc...), FinalLightCurve_HAT-P-18 b_2025-04-30.png (24626331101a...), FinalLightCurve_HAT-P-18 b_2025-04-30.csv (bbda6cc34b3d...)

Compute resources

Wall time 222.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 01:09Z.